PENETRATION TESTING REPORT

Reconnaissance, OSINT & Network Scanning Phases

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Field	Detail
Pentester Name	Taha Chebil
Program / Batch	B084 – NetworkWalks Cybersecurity Internship
Date	11 November 2026
Modules Completed	W2-PM1: Footprinting with Multiple Kali Tools W2-PM2: Google Hacking Database and Google Dorking W2-PM3: Maltego OSINT and Graph Analysis W2-PM4: theHarvester OSINT Reconnaissance W2-PM5: Network Scanning with Zenmap
Client / Target	1. NetworkWalks public website 2. Public search-engine results and educational OSINT sources 3. Microsoft public domain (theHarvester) 4. Local Wi-Fi network used for the lab
Authorization	Educational scope defined by the internship.
Phases Covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

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1. Liability Disclaimer

This report documents cybersecurity activities performed for educational purposes during the NetworkWalks Cybersecurity & Ethical Hacking Internship.

The activities were limited to reconnaissance, public-source research, DNS enumeration, OSINT graph analysis and basic local network host discovery.

The public domains researched included `networkwalks.com` and `microsoft.com`. Search-engine examples included public educational PDF resources and camera-related search results; no authorization to access private surveillance systems is implied. The network scanning exercise was performed on a local network used for training.

No exploitation, unauthorized access, brute-force attack, privilege escalation or denial-of-service activity was performed.

Security testing must only be conducted on systems owned by the tester or covered by valid authorization.

The sample permission letter provided with the internship materials is not presented as authorization issued specifically to the author.

Private network addresses are partially masked in this report. The last octet of IPv4 addresses and the last three bytes of MAC addresses are hidden.

2. Introduction

This report presents the activities completed during Week 2 of my NetworkWalks Cybersecurity & Ethical Hacking Internship, Batch B084.

The project covers two important phases of penetration testing:

- **Phase 1 – Reconnaissance and Footprinting:** Collecting information about a target using publicly available sources and lightweight technical queries.
- **Phase 2 – Network Scanning:** Discovering active hosts and collecting basic information about devices on a local network.

For the first module, W2-PM1, I used six Kali Linux tools to examine the NetworkWalks domain.

For W2-PM2, I practiced Google Dorking to document publicly indexed camera-related pages and mathematics PDF resources. For W2-PM3, I used Maltego to visualize domain, email and public web relationships. For W2-PM4, I used theHarvester to compare single-source and multi-source OSINT results.

For W2-PM5, I used Zenmap on Windows to perform a Ping Scan on my local network, identify responding hosts, and generate a network topology.

This report includes the commands used, actual results, security observations, risk analysis, recommendations and screenshots collected as evidence.

The purpose was to understand how security professionals collect and document information during the early phases of a security assessment.

3. Tools Used

The following tools were used during the Week 2 practical activities.

Tool	Purpose
Kali Linux	Operating system used for footprinting and reconnaissance.
WHOIS	Retrieve domain registration details, registrar, dates and name servers.
WhatWeb	Identify web server software, CMS, plugins and other technologies.
NSLookup	Resolve a domain name to an IP address using DNS.
cURL	Retrieve and analyze HTTP response headers.
WAFW00F	Detect possible Web Application Firewall technologies.
DNSRecon	Enumerate DNS records, mail servers, name servers and DNS-related services.
Zenmap / Nmap	Perform local network host discovery and visualize network topology.
Windows CMD	Identify local IP configuration and MAC address information.

Table 1: Tools used during Week 2

4. Activities Performed

4.1 Footprinting & Reconnaissance (W2-PM1)

I performed reconnaissance against `networkwalks.com` using six Kali Linux tools.

Each tool was used to collect a different type of information about the target.

Task 1 – WHOIS

I first used WHOIS to retrieve publicly available domain registration information.

Command:

```
whois networkwalks.com
```

Results:

- Registrar: GoDaddy.
- Creation date: November 2019.
- Expiration date: November 2027.
- Name servers: ns6135.hostgator.com and ns6136.hostgator.com.
- Registrant details protected by a privacy service.
- DNSSEC: unsigned.

Observation:

WHOIS revealed domain registration information without directly scanning the target web server.

This information can help identify domain management and DNS providers, but it does not necessarily reveal the actual hosting provider.

Task 2 – WhatWeb

Next, I used WhatWeb to identify technologies used by the website.

Command:

```
whatweb https://networkwalks.com
```

Results:

- HTTP status: 200 OK.
- Web server: Apache.
- CMS: WordPress 7.1.4 (reported).
- WordPress Download Manager 3.3.76.
- jQuery 3.7.1.
- Website IP: 192.232.216.X.
- Website title: Networkwalks Academy.
- HttpOnly cookie detected.

Observation:

WhatWeb identified several components of the public website technology stack.

The detected software versions were reported by the tool and were not independently verified.

Technology identification alone does not prove the existence of a vulnerability.

Task 3 – NSLookup

I used NSLookup to resolve the target domain name to an IPv4 address.

Command:

```
nslookup networkwalks.com
```

Results:

- DNS resolver: Google Public DNS.
- DNS server: 8.8.8.8.
- DNS port: 53.
- Resolved IPv4 address: 192.232.216.X.
- Response: Non-authoritative.

Observation:

The domain was successfully resolved.

The returned IPv4 address matched the result obtained using WhatWeb.

This demonstrated how different reconnaissance tools can confirm the same technical information.

Task 4 – cURL

I used cURL to inspect HTTP response headers returned by the website.

Command:

```
curl -I https://networkwalks.com
```

Results:

- HTTP/2 200 OK.
- Server: Apache.
- Content type: HTML, UTF-8.
- WordPress REST API reference: `/wp-json/`.
- Secure and HttpOnly cookie attributes.
- Permissions-Policy header.
- Referrer-Policy header.

Observation:

The response headers provided information about the server, application and browser security configuration.

Some additional security headers were not observed in the captured response and may warrant a configuration review.

Task 5 – WAFW00F

I used WAFW00F to investigate whether a Web Application Firewall was protecting the website.

Command:

```
wafw00f networkwalks.com
```

Results:

- WAF fingerprint: ModSecurity (SpiderLabs).
- WAFW00F version: 2.4.2.
- Number of requests: 2.
- Positive detection result.

Observation:

The tool detected characteristics consistent with ModSecurity.

A WAF helps inspect and filter suspicious web requests.

The detection does not confirm the exact WAF configuration or its effectiveness.

Task 6 – DNSRecon

Finally, I used DNSRecon to enumerate DNS records associated with the target domain.

Command:

```
dnsrecon -d networkwalks.com
```

Results:

- A record: 192.232.216.X.
- Name servers: ns6135.hostgator.com and ns6136.hostgator.com.
- Mail server: mail.networkwalks.com (192.232.216.X).
- DNS server fingerprint: BIND 9.16.23-RH.
- SPF TXT record discovered.
- Google site verification record.
- Eight email autodiscovery SRV answers.

Difficulty encountered:

DNSRecon returned the following message:

```
ERROR No answer for DNSSEC query
```

However, the tool continued and completed the enumeration.

This result was consistent with the unsigned DNSSEC status previously observed using WHOIS.

Observation:

DNSRecon provided additional information about the public DNS infrastructure and email services.

PM1 Summary

All six footprinting tasks were completed successfully.

The collected information included domain registration records, website technologies, DNS resolution results, HTTP response headers, WAF fingerprints and DNS records.

No vulnerability exploitation was performed.

4.2 Google Hacking Database and Google Dorking (W2-PM2)

Objective: Practice advanced search operators to identify publicly indexed resources and understand how search-engine indexing can unintentionally expose information. This was an educational OSINT exercise, not an attempt to gain access to restricted systems.

Task 1 – Camera-related Search Results

I documented ten camera-related results using search operators and public webcam directories. Example searches included:

```
intitle:"webcamXP" inurl:8080
site:skylinewebcams.com "live webcam"
```

Results: Ten camera-related listings or interfaces were recorded in the W2-PM2 exercise table. Some results were public tourist or street webcams; others appeared to be directly reachable camera interfaces. Public reachability does not prove a vulnerability, ownership, authorization, or the absence of access controls. To protect privacy, direct endpoints to potentially private cameras are not reproduced in this public report.

Observation: Search engines can index camera interface pages, but identifying an exposed URL is not equivalent to proving unauthorized access or a security flaw. No credential attempts, control actions, or exploitation were performed.

Task 2 – Mathematics PDF Resources

I used Google Dorking to identify ten mathematics PDF documents or listings. The searches included:

```
intitle:index.of "parent directory" mathematics pdf
site:mit.edu filetype:pdf mathematics
site:edu filetype:pdf "probability theory" lecture notes
```

Results: Ten results were documented, including PDF resources from MIT, Cornell, Stanford, Harvard, UC Davis and an online mathematics library. The findings included lecture notes, mathematics-for-computer-science materials and algebra or probability texts. A PDF file appearing in search results is not necessarily a directory listing; both types were recorded according to the search results.

Observation: Search operators such as `site:`, `filetype:` and `intitle:` improve search precision. Public educational resources can be legitimate and intentionally indexed; indexing alone is not a vulnerability.

PM2 Summary

Both tasks were completed and the two ten-entry evidence tables were prepared. The main lesson was to distinguish public indexing, directory listing, access control and actual vulnerability validation.

4.3 Footprinting and Relationship Mapping with Maltego (W2-PM3)

Objective: Use graphical OSINT transforms to identify and visualize publicly associated entities and relationships.

Task 1 – Install and Configure Maltego

I installed and opened Maltego Graph Desktop 4.14.0 on Windows. I created a new graph, added a Domain entity and set its value to `networkwalks.com`.

Task 2 – Domain and Email Reconnaissance

I ran an email-related transform on the domain entity. The graph displayed the publicly associated address `info@networkwalks.com`. I also explored a separate Person entity and its public website relationships to understand how Maltego visualizes linked OSINT data.

Results:

- A domain-to-email relationship was displayed for `networkwalks.com`.
- A separate graph illustrated person-to-website relationships, including publicly indexed web platforms.
- The graph provided a visual representation of transform output, not proof of ownership or verified identity.

Observation: Maltego helps analysts organize publicly available relationships. A transform result may require independent validation before being treated as confirmed intelligence.

PM3 Summary

The two practical tasks were completed: installation/configuration and basic graph-based reconnaissance. Screenshots of both graphs were saved for evidence.

4.4 OSINT Reconnaissance with theHarvester (W2-PM4)

Objective: Compare search-engine-based and multi-source OSINT reconnaissance using theHarvester on Kali Linux.

Task 1 – Tool Verification and Baidu Search

I verified the installation and usage options of theHarvester 4.10.1:

```
theHarvester -h
```

Then I ran a Baidu-only search:

```
theHarvester -d microsoft.com -l 1000 -b baidu
```

Results: The tool executed and completed without reporting IP addresses, email addresses, people or hosts from the selected Baidu source. An empty result is a valid tool outcome and does not mean the domain has no public infrastructure.

Task 2 – Multi-source Search

I compared the result with a limited multi-source search:

```
theHarvester -d microsoft.com -l 50 -b all
```

Results:

- The output showed numerous Microsoft-related public hostnames and subdomains.

- Hudson Rock processing reported 44 hosts and no IP addresses or emails in its own result summary.
- A DNS fallback reported 14 subdomains.
- Several providers returned missing API-key or authentication warnings, including BuiltWith and other restricted sources.
- Some provider results failed or were unavailable; therefore the multi-source run was not comprehensive.

Difficulties and interpretation: Missing API credentials are source limitations rather than evidence of target misconfiguration. Counts reported by separate providers can overlap and should not be added together as unique hosts. Publicly indexed hostnames are reconnaissance observations, not confirmed vulnerabilities.

PM4 Summary

Both tasks were completed. The exercise showed why results vary by source and why analyst reports must record failures, rate limits and API dependencies alongside successful findings.

4.5 Network Scanning with Zenmap (W2-PM5)

For the network scanning activity, I used Zenmap to perform host discovery on a local Wi-Fi network.

The objective was to identify the network configuration, discover responding devices, collect IP and MAC addresses and generate a network topology.

Task 1 – Install Zenmap

I downloaded and installed Zenmap on my Windows computer.

The application opened successfully, confirming that it was ready for network scanning.

Task 2 – Identify Local Network

I used Windows Command Prompt to identify the active Wi-Fi adapter and its network configuration.

Command:

```
ipconfig
```

Results:

- Network adapter: Wi-Fi.
- IPv4 address: 192.168.X.X.
- Subnet mask: 255.255.255.0.
- Default gateway: 192.168.X.X.

- Local subnet: 192.168.X.X/24.

The /24 subnet was used for the host discovery exercise.

The last octet of each address has been hidden for privacy.

Task 3 – Discover Live Hosts

I opened Zenmap, entered the local subnet, and selected the Ping Scan profile.

Command:

```
nmap -sn 192.168.X.X/24
```

Note: The command above is a redacted representation of the actual scan command.

The `-sn` option performs host discovery without a port scan.

Results:

- 256 IPv4 addresses scanned.
- 3 responding hosts detected.
- Scan duration: 3.30 seconds.
- Scan completed successfully.

Task 4 – Count Live Hosts

The Zenmap output reported:

3 Hosts Up

This represents the number of hosts that responded to the discovery scan.

Devices that do not respond may still be connected to the network.

Task 5 – Identify IP Addresses

The following hosts were discovered during the scan.

Host	IP Address	Status
Host 1	192.168.X.X	Up
Host 2	192.168.X.X	Up
Host 3	192.168.X.X	Up

Table 2: Discovered hosts with partially masked IP addresses

Host 1 corresponded to the default gateway.

Host 3 corresponded to my Windows computer.

The exact identity of Host 2 was not confirmed.

Task 6 – Identify MAC Addresses

Zenmap displayed MAC addresses for two neighboring hosts.

I used the following command to identify the MAC address of my Windows Wi-Fi adapter:

```
ipconfig /all
```

Host	MAC Address
Host 1	98:X:X:XX:XX:XX
Host 2	98:X:X:XX:XX:XX
Host 3	04:X:X:XX:XX:XX

Table 3: MAC addresses with the last three bytes hidden

Observation:

MAC addresses identify network interfaces within a local network.

The first three bytes may be used for vendor identification.

The last three bytes were masked for privacy.

Task 7 – Network Topology

After completing the Ping Scan, I opened the Topology tab in Zenmap.

The graphical view displayed the three responding hosts.

I then exported the topology as a PDF file.

Observation:

The topology provides a visual representation of discovered network hosts.

It does not necessarily represent the physical connections between devices.

PM5 Summary

The Zenmap exercise was completed successfully.

Three responding hosts were discovered, their IP and MAC information was collected, and the network topology was exported as a PDF.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning exercises, I identified the following security observations.

The risk levels below are preliminary review priorities, not validated vulnerability ratings.

#	Finding	Potential Impact	Review Priority
1	Web technologies identifiable	Software fingerprints may assist further reconnaissance.	Medium
2	Server IP identifiable	Public DNS reveals the website endpoint.	Low
3	HTTP technical information	Headers and REST API references reveal application details.	Low
4	WAF technology identifiable	WAF fingerprint reveals part of the defensive stack.	Low
5	DNS information exposed	DNS records describe public infrastructure and email services.	Low
6	DNSSEC unsigned	DNS responses lack DNSSEC authentication at the domain delegation level.	Medium
7	Multiple local hosts visible	Unexpected devices would require administrator review.	Low

Table 4: Preliminary risk analysis

Important note:

The findings above are observations from reconnaissance and host discovery.

They do not establish confirmed exploitable vulnerabilities.

The presence of a software version, IP address, DNS record, REST API endpoint or WAF does not automatically mean that a system is insecure.

Further authorized testing would be necessary to validate any suspected vulnerability.

Additional OSINT Observations (PM2–PM4)

- **Public camera interfaces:** Search results can include private-looking endpoints. Ownership and access controls were not verified; do not classify these as confirmed vulnerabilities. *Review priority: contextual / requires authorization.*
- **Public PDF resources:** The educational documents found were publicly indexed. Their existence alone is not a security weakness. *Review priority: informational.*
- **Maltego entity relationships:** Public associations can be useful leads but may be incomplete or incorrect. *Review priority: informational.*
- **theHarvester subdomains:** Hostname discovery helps build an inventory but does not establish exploitability. *Review priority: informational.*
- **OSINT provider errors:** Missing API keys and limited source coverage affect the completeness of findings. *Review priority: methodology / data quality.*

6. Recommendations

Based on the observations, I recommend the following security improvements.

1. **Review exposed technology information.**

   Regularly review which CMS, plugin and server details are publicly detectable.

2. **Keep software updated.**

   Maintain supported versions of WordPress, plugins, JavaScript libraries and web server software.

3. **Review HTTP security headers.**

   Evaluate appropriate Content-Security-Policy, Strict-Transport-Security and X-Content-Type-Options settings.

4. **Review DNS records regularly.**

   Remove obsolete DNS records and verify that public records match the intended configuration.

5. **Evaluate DNSSEC.**

   Consider enabling DNSSEC with the domain registrar and authoritative DNS provider.

6. **Maintain the WAF.**

   Review ModSecurity configuration, update applicable rules and monitor security logs.

7. **Perform regular internal network discovery.**

   Use authorized network scans to identify responding devices and maintain an asset inventory.

8. **Investigate unknown devices.**

   Verify the identity and authorization of unexpected hosts.

9. **Maintain network documentation.**

   Keep network inventories and topology diagrams updated.

10. **Respect authorization and privacy.**

    Perform testing only within an approved scope and redact sensitive data before public publication.

11. **Review public search-engine exposure.**

    Asset owners should periodically review indexed pages, public directory listings and camera interfaces. Restrict non-public content through proper authentication and network controls rather than relying on search-engine removal alone.

12. **Validate OSINT findings before escalating.**

    Cross-check public search results and Maltego or theHarvester transforms; document source coverage, timestamps and API failures.

7. Conclusion

During Week 2 of my NetworkWalks Cybersecurity & Ethical Hacking Internship, I completed practical exercises covering reconnaissance, footprinting and network scanning.

In W2-PM1, I used six Kali Linux tools to collect information about the NetworkWalks domain.

WHOIS provided registration details, WhatWeb identified web technologies, NSLookup resolved the domain, cURL displayed HTTP headers, WAFW00F detected a WAF fingerprint, and DNSRecon enumerated DNS records.

In W2-PM2, I documented ten camera-related search results and ten mathematics PDF resources, learning the distinction between publicly indexed content and a validated vulnerability.

In W2-PM3, I used Maltego to map a domain-to-email relationship and explore graphical connections between people and public websites.

In W2-PM4, I used theHarvester to compare a Baidu-only search with multi-source OSINT. The latter produced public hostnames while also exposing API authentication and provider-coverage limitations.

In W2-PM5, I used Zenmap to discover active hosts on a local network.

The Ping Scan identified three responding hosts out of 256 scanned addresses.

I also identified IP and MAC address information and exported a network topology.

These exercises demonstrated the importance of information gathering during the early phases of a penetration test.

I learned that a professional cybersecurity report should explain the commands performed, the results observed, their security significance and appropriate recommendations.

The activities were limited to reconnaissance and scanning. No exploitation was performed.

8. Evidences Collected

The following screenshots were collected during the Week 2 practical activities.

Private addresses and MAC addresses should be partially masked in the actual screenshot files before public publication.

8.1 W2-PM1 – Footprinting Evidence

Evidence 1 – WHOIS



Figure 1: WHOIS domain registration results

Evidence 2 – WhatWeb

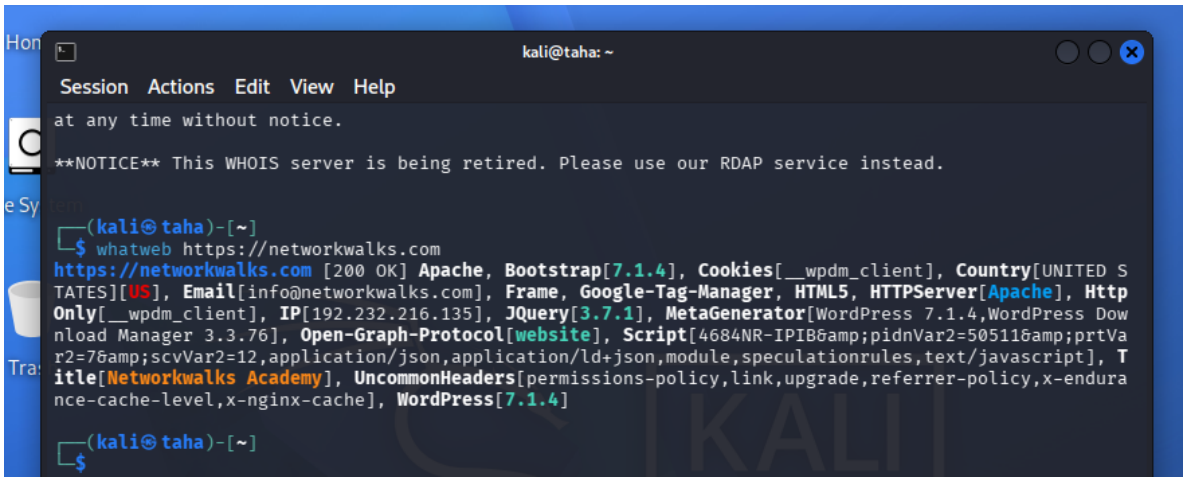


Figure 2: WhatWeb website technology fingerprinting

Evidence 3 – NSLookup

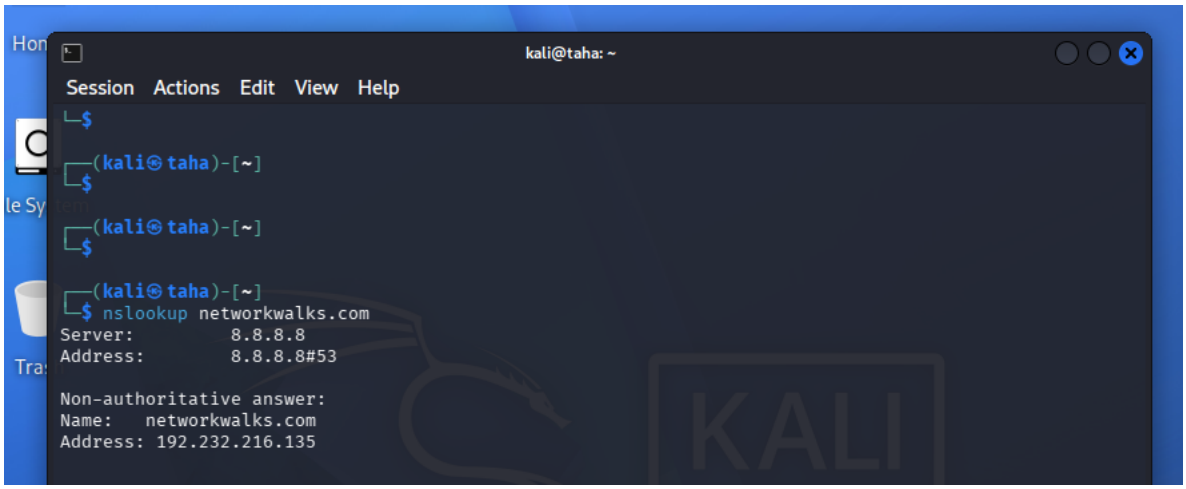


Figure 3: NSLookup domain resolution results

Evidence 4 – cURL

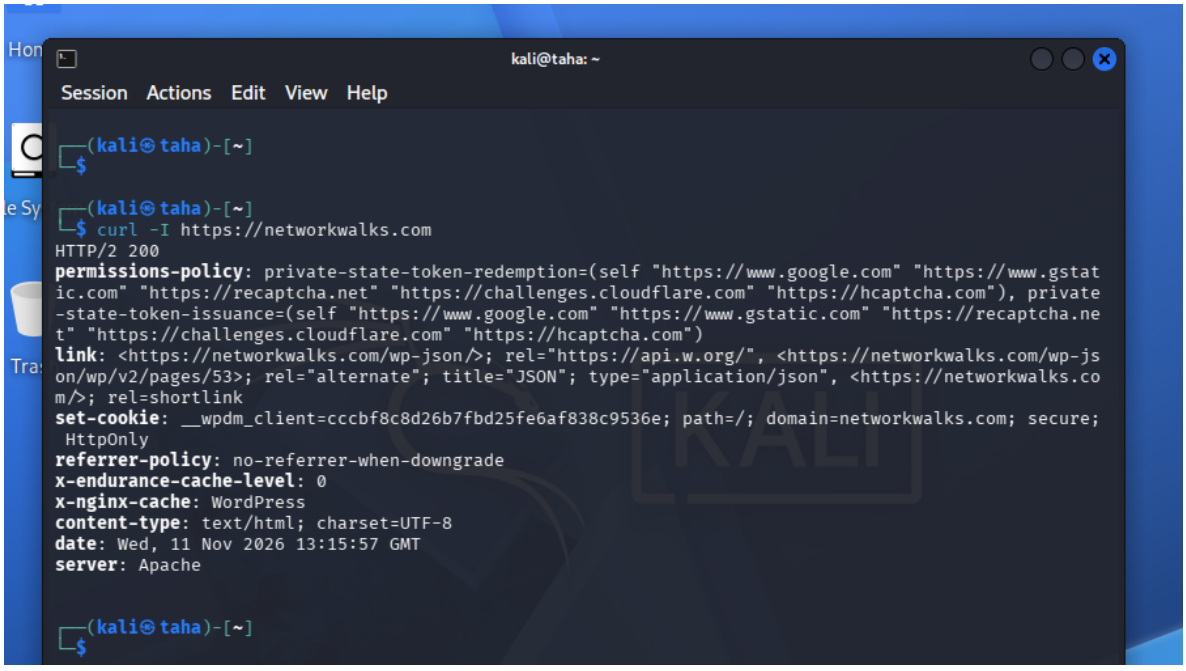


Figure 4: cURL HTTP response headers

Evidence 5 – WAFW00F

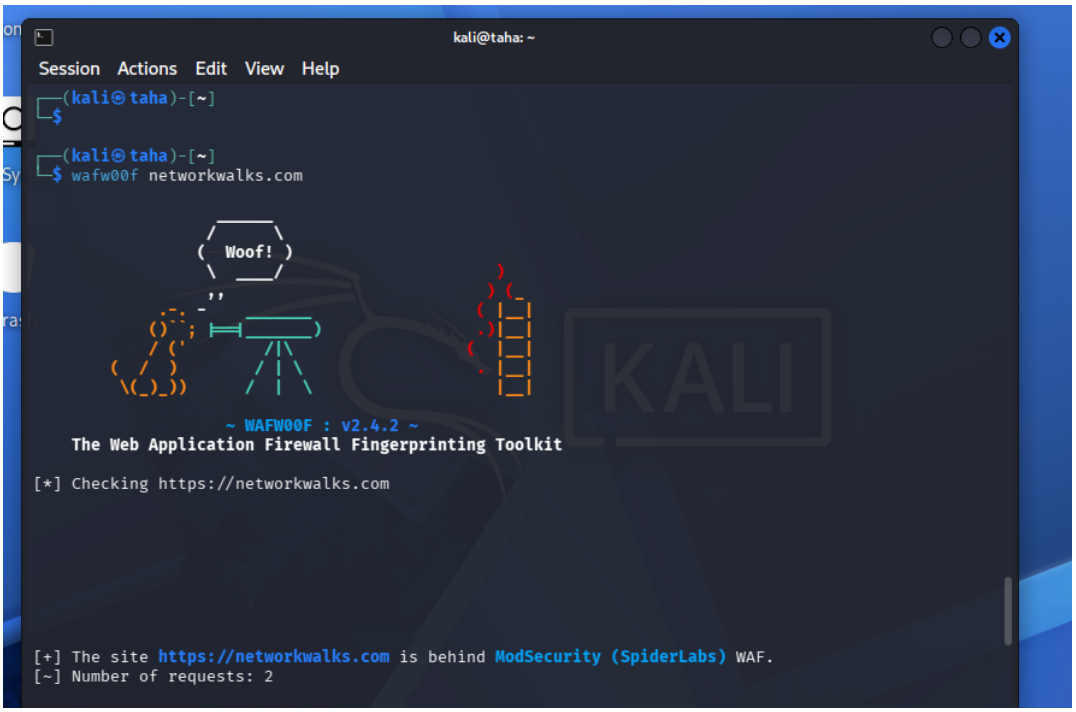


Figure 5: WAFW00F ModSecurity detection

Evidence 6 – DNSRecon

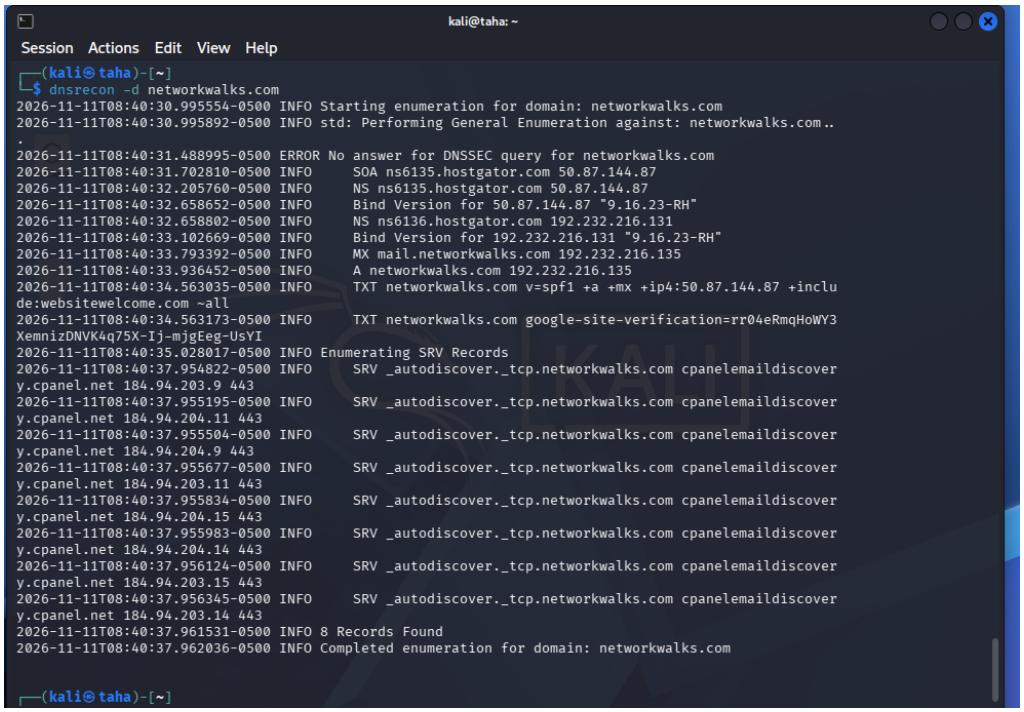


Figure 6: DNSRecon DNS enumeration results

8.2 W2-PM2 – Google Dorking Evidence

Evidence: The ten-entry camera-related results table and ten-entry mathematics PDF table are documented in the separate `W2-PM2_GHDB_TASKS_Report.pdf`. Direct private camera endpoints are intentionally omitted from this public report.

8.3 W2-PM3 – Maltego Evidence

Evidence – Domain and Email Graph

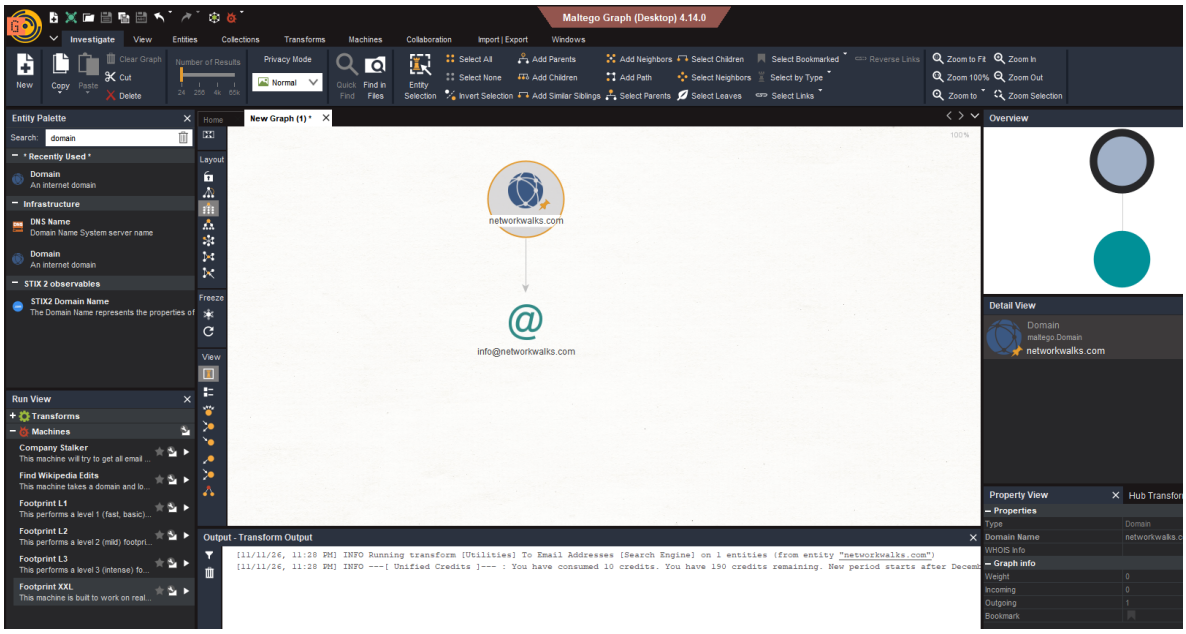


Figure 7: Maltego graph showing domain-to-email relationship

Evidence – Person and Website Graph

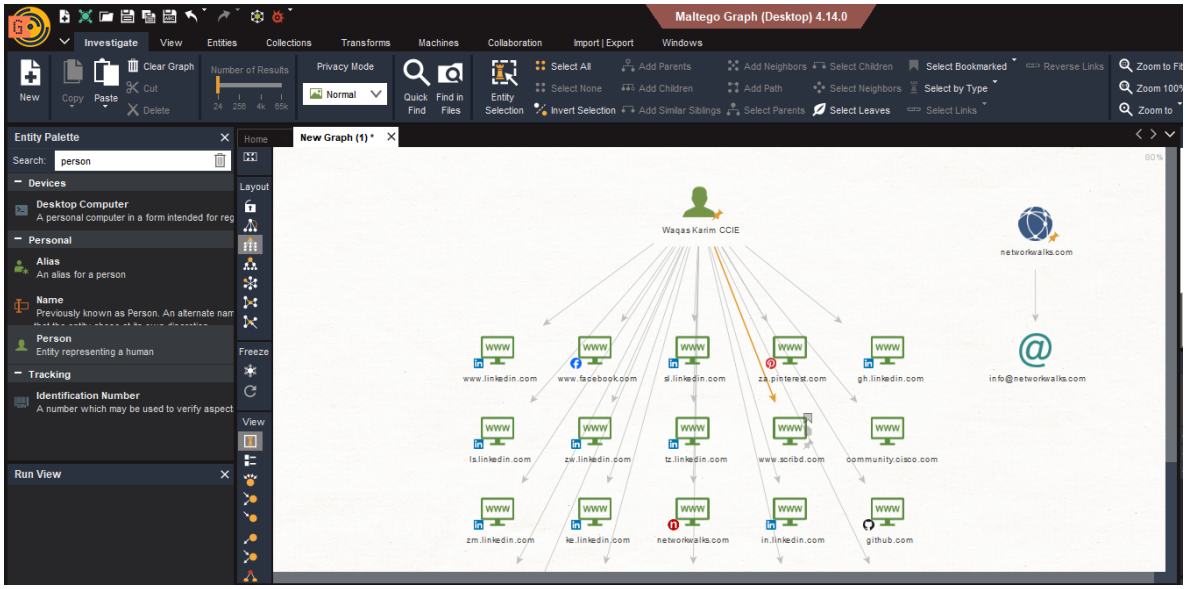


Figure 8: Maltego visualization of public person-to-website relationships

8.4 W2-PM4 – theHarvester Evidence

Evidence – Tool Help and Version

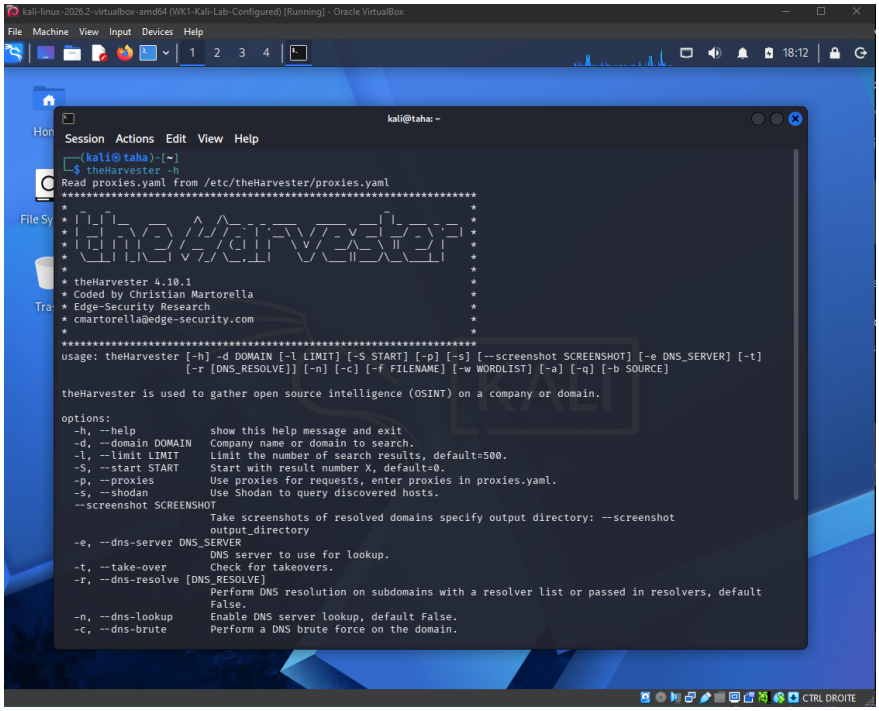


Figure 9: theHarvester 4.10.1 help and available options

Evidence – Baidu-only Search

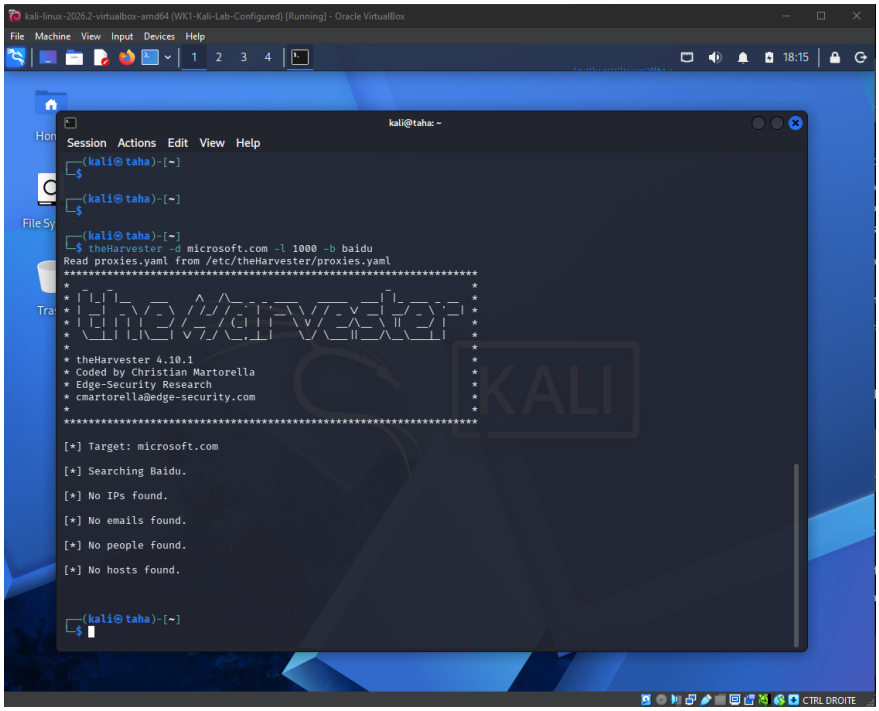


Figure 10: Baidu source completed with no results

Evidence – Multi-source API Warnings

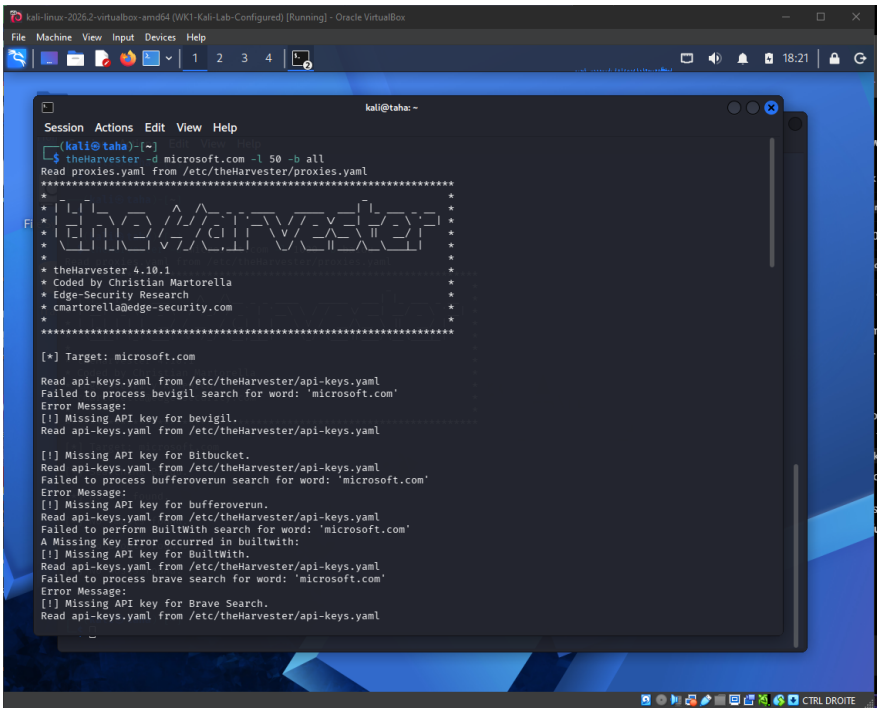


Figure 11: Multi-source search and API authentication limitations

Evidence – Hudson Rock and DNS Fallback

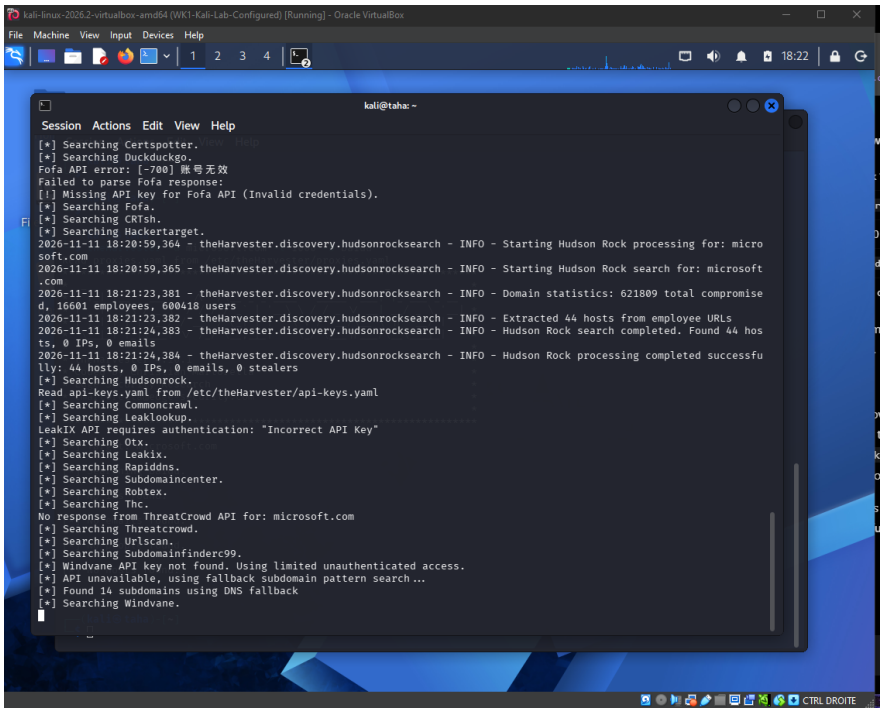


Figure 12: Hudson Rock host findings and DNS fallback output

Evidence – Final Host Output

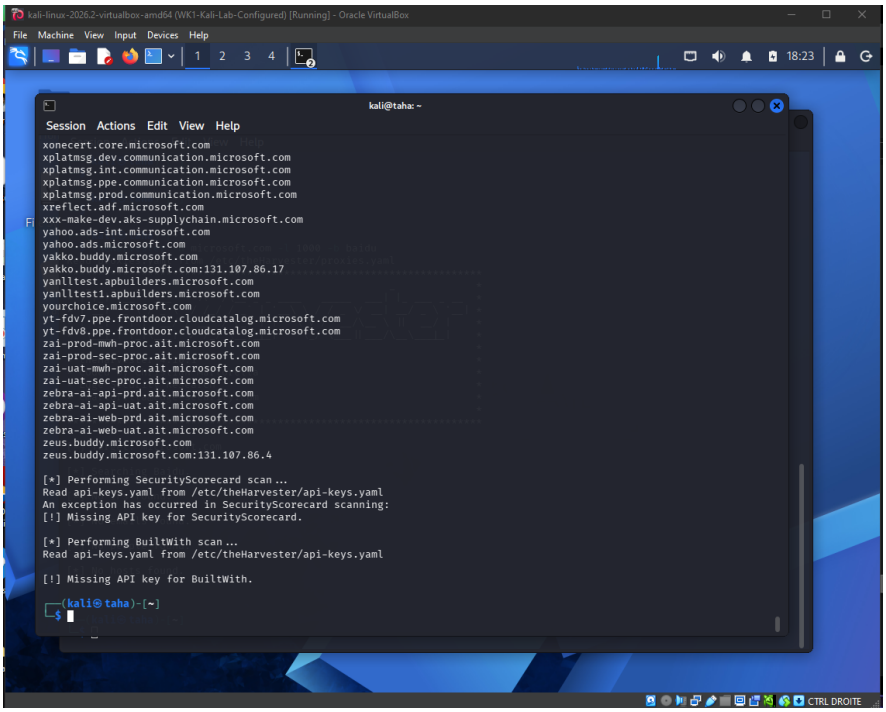


Figure 13: theHarvester final results and additional provider warnings

8.5 W2-PM5 – Network Scanning Evidence

Evidence 7 – Zenmap Installation

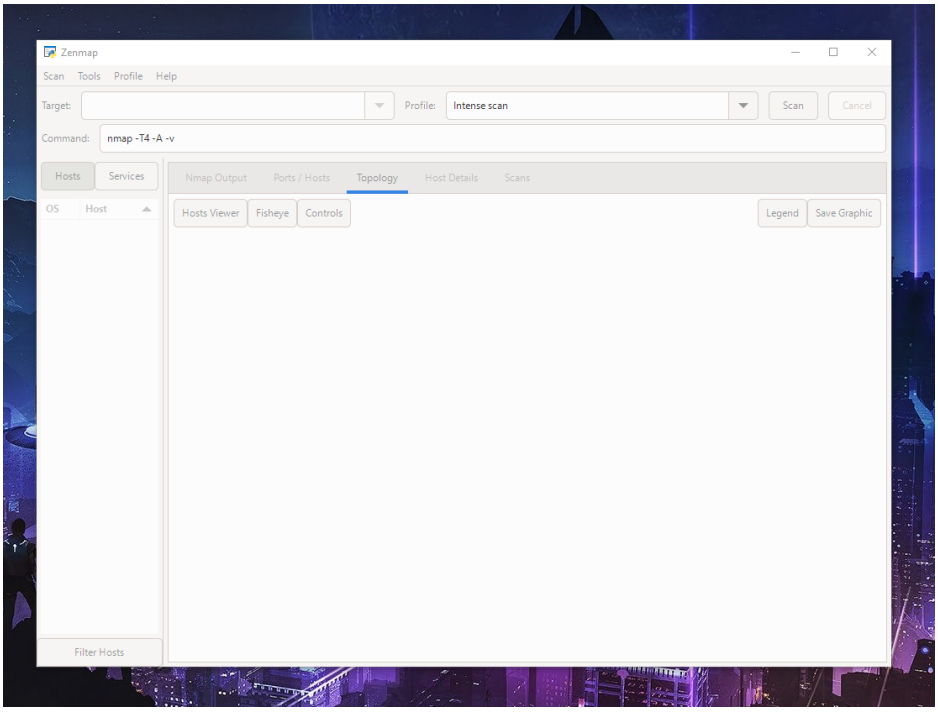


Figure 14: Zenmap application installed on Windows

Evidence 8 – Local Network Configuration

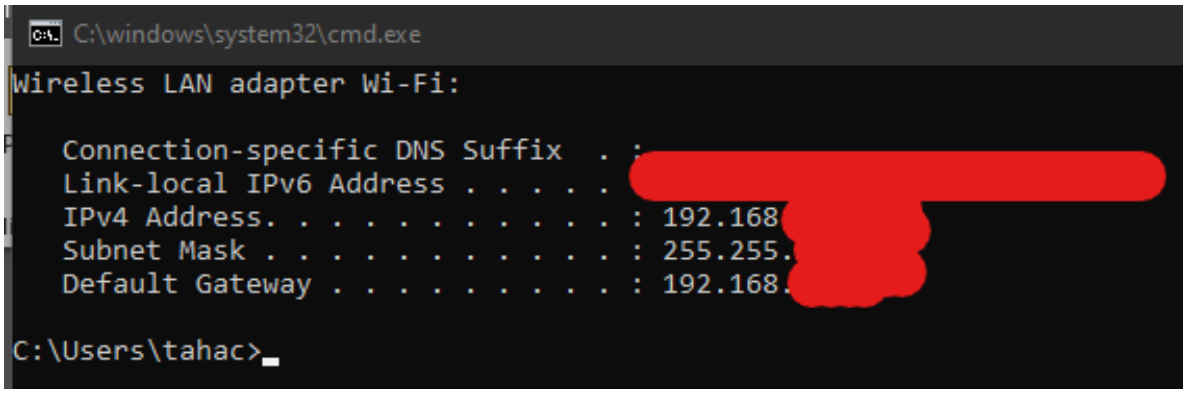


Figure 15: Windows network configuration

Evidence 9 – Ping Scan

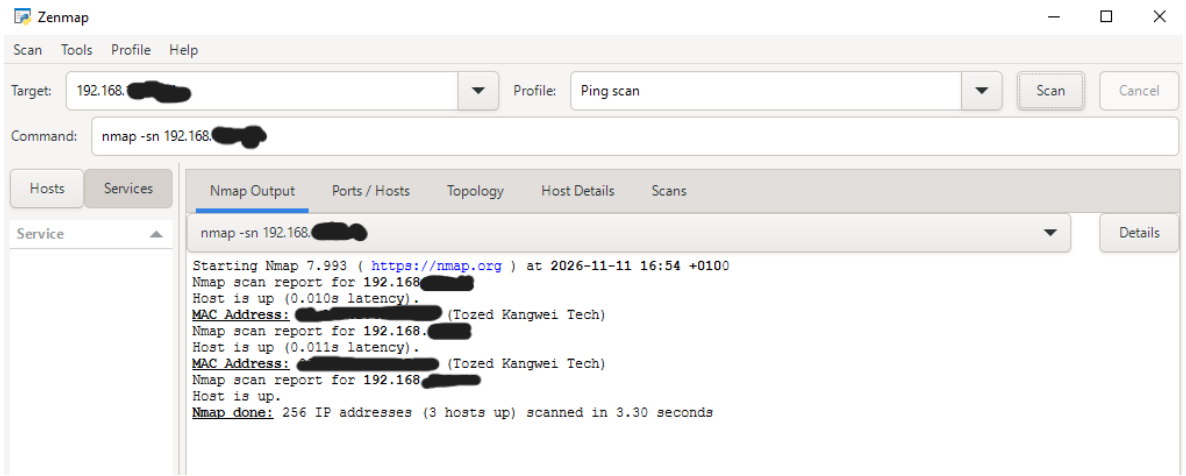


Figure 16: Zenmap Ping Scan and responding hosts

Evidence 10 – Network Topology

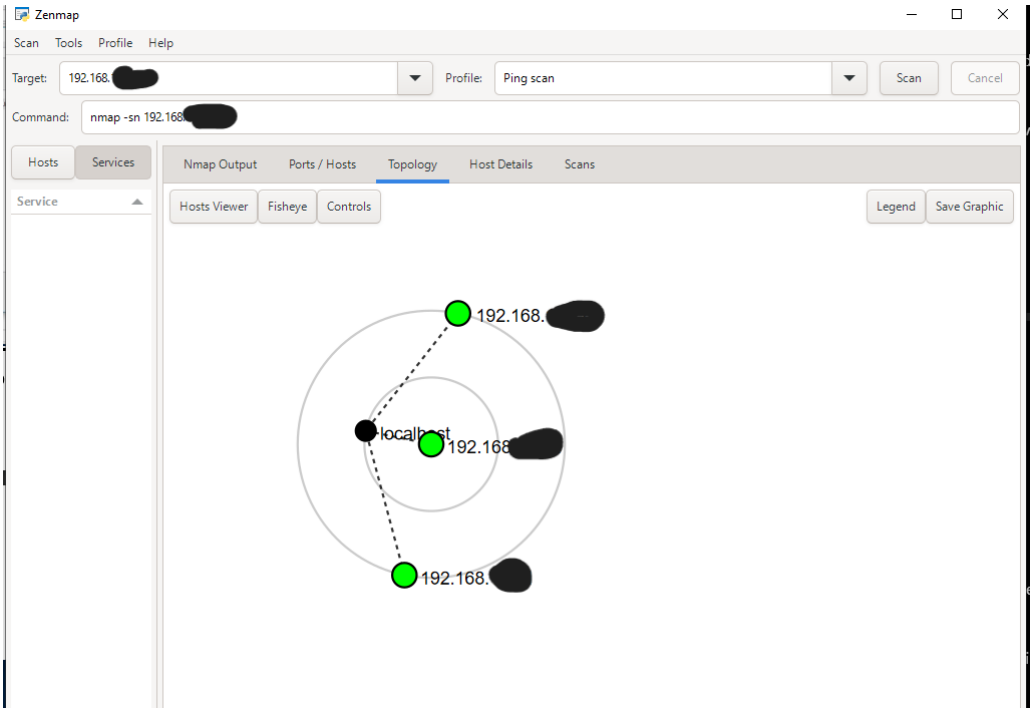


Figure 17: Zenmap network topology

End of Report

Taha Chebil – Batch B084

NetworkWalks Cybersecurity Internship

November 2026